

Measure Theory – 3

Independence of Events, Events Occurring Infinitely Often,
Measurable Functions and Piecewise Maps,
Induced Measures, and Composition of Random Variables

Ravleen Bajaj

1 Independence of Events

1.1 Definition and Complements

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, $\mathcal{B}_{\mathbb{R}}$ is the Borel σ -algebra on $\mathbb{R}$, and λ is Lebesgue measure.

Definition 1.1 (Mutual independence). Events $A_1, \dots, A_m \in \mathcal{F}$ are **mutually independent** if for every non-empty $I \subseteq \{1, \dots, m\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

Lemma 1.2 (Complements of independent events). *If $A_1, \dots, A_m$ are mutually independent, then so are $A_1^c, \dots, A_m^c$.*

Proof. It suffices to replace one event at a time. Fix I containing 1, put $J = I \setminus \{1\}$ and $C = \bigcap_{i \in J} A_i$ (with $C = \Omega$ if $J = \emptyset$). Since $C = (A_1 \cap C) \sqcup (A_1^c \cap C)$,

$$\mathbb{P}(A_1^c \cap C) = \mathbb{P}(C) - \mathbb{P}(A_1) \mathbb{P}(C) = \mathbb{P}(A_1^c) \prod_{i \in J} \mathbb{P}(A_i).$$

So $A_1^c, A_2, \dots, A_m$ are mutually independent; repeat for each index. □

1.2 Problem: Probability of a Finite Union

Let $A_1, \dots, A_m$ be mutually independent events in $(\Omega, \mathcal{F}, \mathbb{P})$.

Key Result.

$$\mathbb{P}\left(\bigcup_{i=1}^m A_i\right) = 1 - \prod_{i=1}^m (1 - \mathbb{P}(A_i)).$$

Proof. 1. **De Morgan.** $(\bigcup_{i=1}^m A_i)^c = \bigcap_{i=1}^m A_i^c$.

2. **Complement rule.** Using $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$,

$$\mathbb{P}\left(\bigcup_{i=1}^m A_i\right) = 1 - \mathbb{P}\left(\bigcap_{i=1}^m A_i^c\right).$$

3. **Independence.** By Lemma 1.2,

$$\mathbb{P}\left(\bigcap_{i=1}^m A_i^c\right) = \prod_{i=1}^m \mathbb{P}(A_i^c) = \prod_{i=1}^m (1 - \mathbb{P}(A_i)).$$

4. **Substitute** Step 3 into Step 2. □

Example 1.3. Roll a fair die m times; let $A_i = \{\text{six on roll } i\}$. Then $\mathbb{P}(\text{at least one six}) = 1 - (5/6)^m$, which is $1 - 625/1296 \approx 0.518$ for $m = 4$. If $\mathbb{P}(A_i) \geq p > 0$ for all i , the union has probability at least $1 - (1 - p)^m \rightarrow 1$.

Remark. • Pairwise independence is *not* enough: the product rule must hold for every subfamily.

- The formula $\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i)$ requires *disjoint* events, not independent ones.

2 Events Occurring Infinitely Often

2.1 Definitions

For events $\{A_n\}_{n \geq 1}$ in $(\Omega, \mathcal{F}, \mathbb{P})$:

$$\begin{aligned} \{A_n \text{ i.o.}\} &= \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k, \\ \liminf_{n \rightarrow \infty} A_n &= \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k. \end{aligned}$$

- $\omega \in \{A_n \text{ i.o.}\}$ iff ω lies in **infinitely many** A_n .
- $\omega \in \liminf A_n$ iff ω lies in **all but finitely many** A_n .

Both sets are countable unions/intersections of events, hence belong to $\mathcal{F}$. Also $\liminf A_n \subseteq \limsup A_n$, and this inclusion can be strict.

2.2 Problem: A Lower Bound on $\mathbb{P}(A_n \text{ i.o.})$

Put $B_n = \bigcup_{k \geq n} A_k$. Then $B_1 \supseteq B_2 \supseteq \cdots$ and $\limsup A_n = \bigcap_n B_n$.

Key Result.

If there exist $\epsilon > 0$ and N with $\mathbb{P}(A_n) \geq \epsilon$ for all $n \geq N$, then

$$\mathbb{P}(A_n \text{ i.o.}) \geq \epsilon.$$

Proof. For $n \geq N$, $A_n \subseteq B_n$, so $\mathbb{P}(B_n) \geq \mathbb{P}(A_n) \geq \epsilon$. Since $\mathbb{P}$ is finite and $B_n \downarrow \limsup A_n$, continuity from above gives

$$\mathbb{P}(A_n \text{ i.o.}) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n) \geq \epsilon. \quad \square$$

The same argument gives the general inequality

$$\mathbb{P}(\liminf A_n) \leq \liminf_n \mathbb{P}(A_n) \leq \limsup_n \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n).$$

2.3 Borel–Cantelli Lemmas

Theorem 2.1 (First Borel–Cantelli). *If $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*

Proof. For each n , $\mathbb{P}(A_n \text{ i.o.}) \leq \mathbb{P}(B_n) \leq \sum_{k \geq n} \mathbb{P}(A_k) \rightarrow 0$, the tail of a convergent series. $\square$

Theorem 2.2 (Second Borel–Cantelli). *If the A_n are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

Proof. By De Morgan, $\{A_n \text{ i.o.}\}^c = \bigcup_n \bigcap_{k \geq n} A_k^c$. For $m \geq n$, independence and $1 - t \leq e^{-t}$ give

$$\mathbb{P}\left(\bigcap_{k=n}^m A_k^c\right) = \prod_{k=n}^m (1 - \mathbb{P}(A_k)) \leq \exp\left(-\sum_{k=n}^m \mathbb{P}(A_k)\right) \xrightarrow{m \rightarrow \infty} 0.$$

So each $\bigcap_{k \geq n} A_k^c$ is null, and so is the countable union. $\square$

Remark. • The lower bound ϵ does *not* give probability 1; that needs independence and a divergent sum (second lemma).

- Independence cannot be dropped: if $A_n = A$ for all n with $\mathbb{P}(A) = \frac{1}{2}$, then $\sum \mathbb{P}(A_n) = \infty$ but $\mathbb{P}(A_n \text{ i.o.}) = \frac{1}{2}$.

3 Measurable Functions and Piecewise Maps

3.1 Definition and Tools

Definition 3.1 (Measurable function). Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A map $f : X \rightarrow Y$ is $(\mathcal{F}, \mathcal{G})$ -**measurable** if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{G}$.

Lemma 3.2 (Generating class). *If $\mathcal{G} = \sigma(\mathcal{C})$, then f is measurable iff $f^{-1}(C) \in \mathcal{F}$ for all $C \in \mathcal{C}$.*

Proof. $\{B : f^{-1}(B) \in \mathcal{F}\}$ is a σ -algebra, since preimages commute with complements and countable unions. If it contains $\mathcal{C}$, it contains $\sigma(\mathcal{C})$. $\square$

Consequently $f : X \rightarrow \mathbb{R}$ is measurable iff $\{f \leq a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$, and every continuous $g : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable.

Lemma 3.3 (Gluing on a measurable partition). *Let $X = E_1 \sqcup \cdots \sqcup E_k$ with each $E_i \in \mathcal{F}$, and suppose each restriction $f|_{E_i}$ is measurable. Then f is measurable.*

Proof. For $B \in \mathcal{G}$, $f^{-1}(B) = \bigcup_{i=1}^k (f|_{E_i})^{-1}(B)$. Each term lies in $\mathcal{F}$ (as $E_i \in \mathcal{F}$), and $\mathcal{F}$ is closed under finite unions. $\square$

3.2 Problem: A Piecewise Real-Valued Function

Consider $f : (\mathbb{R}, \mathcal{B}_{\mathbb{R}}) \rightarrow (\text{Range}(f), \mathcal{B}_{\text{Range}(f)})$,

$$f(x) = \begin{cases} -3x, & x < 0, \\ x, & 0 \leq x \leq 2, \\ 1 - x, & x > 2. \end{cases}$$

Range of each piece.

Piece	Domain	Range
1	$(-\infty, 0)$	$(0, \infty)$
2	$[0, 2]$	$[0, 2]$
3	$(2, \infty)$	$(-\infty, -1)$

Hence $\text{Range}(f) = (-\infty, -1) \cup [0, \infty)$. The first two ranges overlap on $(0, 2]$, so f is not injective.

Measurability. The intervals $(-\infty, 0)$, $[0, 2]$, $(2, \infty)$ are Borel and partition $\mathbb{R}$, and each piece is continuous, hence Borel. By Lemma 3.3, f is measurable. Explicitly, for Borel B ,

$$f^{-1}(B) = \left(-\frac{1}{3}(B \cap (0, \infty)) \right) \cup (B \cap [0, 2]) \cup \left(1 - (B \cap (-\infty, -1)) \right),$$

a union of three Borel sets (images of Borel sets under affine maps with non-zero slope).

Example 3.4. For $B = [1, 3]$: piece 1 gives $[-1, -\frac{1}{3}]$, piece 2 gives $[1, 2]$, and piece 3 gives $\emptyset$. So $f^{-1}([1, 3]) = [-1, -\frac{1}{3}] \cup [1, 2]$.

Remark. • Continuity of each piece is only useful because the *domains* of the pieces are themselves Borel sets; check this.

- Invert each piece separately and intersect with its own domain; otherwise preimages from different pieces get mixed up.

4 Induced (Push-Forward) Measures

4.1 Definition

Definition 4.1 (Induced measure). Let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \rightarrow Y$ a measurable map into $(Y, \mathcal{N})$. The **induced (push-forward) measure** is

$$\mu_f(B) = \mu(f^{-1}(B)), \quad B \in \mathcal{N}.$$

Proposition 4.2. μ_f is a measure on $(Y, \mathcal{N})$; if μ is a probability measure, so is μ_f .

Proof. $\mu_f(\emptyset) = 0$. For disjoint $B_j \in \mathcal{N}$, the sets $f^{-1}(B_j)$ are disjoint and $f^{-1}(\bigcup_j B_j) = \bigcup_j f^{-1}(B_j)$, so countable additivity of μ transfers to μ_f . $\square$

Change of variables.

For measurable $g \geq 0$ (or integrable),

$$\int_Y g d\mu_f = \int_X g \circ f d\mu.$$

This holds for $g = \mathbf{1}_B$ by definition, extends to simple functions by linearity, and to general g by monotone convergence.

4.2 Problem: Push-Forward of Lebesgue Measure

Let $\mu = \lambda$ and

$$f(x) = \begin{cases} x, & x < 0, \\ 2x, & x \geq 0. \end{cases}$$

For $B \in \mathcal{B}_{\mathbb{R}}$, $f^{-1}(B) = (B \cap (-\infty, 0)) \cup \frac{1}{2}(B \cap [0, \infty))$. Since λ scales by $\frac{1}{2}$ under $x \mapsto x/2$,

$$\lambda_f(B) = \lambda(B \cap (-\infty, 0)) + \frac{1}{2} \lambda(B \cap [0, \infty)).$$

So λ_f has density 1 on $(-\infty, 0)$ and density $\frac{1}{2}$ on $[0, \infty)$. For instance, $\lambda_f((-1, 2)) = 1 + \frac{1}{2} \cdot 2 = 2$.

Note:

- The push-forward is computed through the *preimage*, never the image: $\mu_f(B) \neq \mu(f(B))$ in general.
- Measurability of f is what ensures $f^{-1}(B) \in \mathcal{M}$ so that μ_f is defined.

5 Random Variables Defined by Cases

5.1 Problem: Gluing Two Random Variables on an Event

Let X, Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ and $A \in \mathcal{F}$. Define

$$Z(\omega) = X(\omega)\mathbf{1}_A(\omega) + Y(\omega)\mathbf{1}_{A^c}(\omega) = \begin{cases} X(\omega), & \omega \in A, \\ Y(\omega), & \omega \in A^c. \end{cases}$$

Key Result.

Z is a random variable.

Proof. For $B \in \mathcal{B}_{\mathbb{R}}$,

$$Z^{-1}(B) = (A \cap X^{-1}(B)) \cup (A^c \cap Y^{-1}(B)).$$

1. X, Y are random variables, so $X^{-1}(B), Y^{-1}(B) \in \mathcal{F}$.
2. $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$.
3. $\mathcal{F}$ is closed under finite intersections and unions, so $Z^{-1}(B) \in \mathcal{F}$. □

Note:

- $A \in \mathcal{F}$ is essential. With $X \equiv 1, Y \equiv 0$ we get $Z = \mathbf{1}_A$, and $Z^{-1}(\{1\}) = A$, which is not an event if $A \notin \mathcal{F}$.
- This is Lemma 3.3 applied to the partition $\{A, A^c\}$ of Ω .

6 Composition of Random Variables and Equality of Law

6.1 Setup

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be a random variable with law $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$, and let $Y : (\mathbb{R}, \mathcal{B}_{\mathbb{R}}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be Borel measurable. Then $Y \circ X$ is a random variable, since

$$(Y \circ X)^{-1}(B) = X^{-1}(Y^{-1}(B)) \in \mathcal{F}.$$

6.2 Problem: Law of $Y \circ X$

Key Result.

Regarding Y as a random variable on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mathbb{P}_X)$, its law equals the law of $Y \circ X$ on $(\Omega, \mathcal{F}, \mathbb{P})$.

Proof. For $B \in \mathcal{B}_{\mathbb{R}}$,

$$\mathbb{P}_X(Y^{-1}(B)) = \mathbb{P}(X^{-1}(Y^{-1}(B))) = \mathbb{P}((Y \circ X)^{-1}(B)) = \mathbb{P}_{Y \circ X}(B).$$

Since B was arbitrary, the two laws coincide. □

Corollary 6.1. For non-negative Borel h ,

$$\mathbb{E}[h(Y(X))] = \int_{\mathbb{R}} h \circ Y \, d\mathbb{P}_X = \int_{\mathbb{R}} h \, d\mathbb{P}_{Y \circ X}.$$

Remark. Probabilities and expectations of functions of X can be computed entirely on $\mathbb{R}$ from the law $\mathbb{P}_X$, without reference to Ω . This is the change of variables formula of Section 4 applied to $f = X$.

- The order of composition matters: $(Y \circ X)^{-1}(B) = X^{-1}(Y^{-1}(B))$, not $Y^{-1}(X^{-1}(B))$.
- Y must be Borel measurable, not merely defined on the range of X ; otherwise $Y^{-1}(B)$ may fail to lie in $\mathcal{B}_{\mathbb{R}}$.

References

- [1] S. Basu, T. Butler, D. Estep, and N. Panda, *A Ramble Through Probability: How I Learned to Stop Worrying and Love Measure Theory*, Computational Science & Engineering, vol. 29, SIAM, Philadelphia, 2024.